

# SSLM4: Achieving Useful Agent Capabilities at GPT-2 Scale

## A Super Small But Usable Architecture

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### Abstract

We present SSLM4, a hybrid decoder-only language model designed to achieve **useful agent-level capabilities at GPT-2 scale ( 1.5B parameters)**. The architecture combines three proven innovations: Multi-head Latent Attention (MLA) from DeepSeek V2/V3 for compressed KV-cache and multi-turn conversation memory efficiency, Mamba2 state-space layers for efficient long-context processing and agent planning, and an aux-loss-free Mixture-of-Experts (MoE) module for parameter-efficient scaling without load-balancing auxiliary losses. SSLM4 v2 achieves **1.45B total parameters** (comparable to GPT-2 XL) with only  $\sim 768\text{M}$  active parameters per forward pass, making it well-suited for efficient agent inference. We present complete training results from 22K vocab (SSLM4 v1) to 53K vocab (SSLM4 v2), alongside evaluations on C-Eval, TruthfulQA, and GSM8K benchmarks.

## 1 Introduction

### 1.1 Goal: Useful Agent at GPT-2 Scale

The original GPT-2 models [Radford et al., 2019] (117M / 345M / 774M / 1.5B) established that decoder-only transformers at the  $\sim 1\text{B}$  parameter scale can exhibit surprising language understanding and generation capabilities. Our goal is to match or exceed GPT-2’s utility while being more parameter-efficient, by combining modern architectural innovations (MLA, Mamba2, MoE) that were not available in 2019.

“Useful agent level” means the model should be capable of:

1. **Tool use:** Calling external tools/APIs in a coherent sequence (ReAct-style) [Yao et al., 2023]
2. **Multi-step reasoning:** Breaking down complex tasks into subgoals and executing them
3. **Long-horizon memory:** Maintaining context across extended conversations ( $\geq 1024$  tokens)
4. **Multilingual tool navigation:** Operating on both English and Chinese inputs
5. **Code generation & execution:** Producing correct code for simple tasks

This goal differentiates DenseSLM4MoE from pure benchmark-optimized models — we prioritize architectural choices that enable agent workflows, not just high C-Eval scores.

### 1.2 Why Hybrid MLA + Mamba2 + MoE?

For agent workloads, three capabilities matter:

Table 1: Architectural solutions for agent capabilities

| Capability           | Requirement                    | Solution                             |
|----------------------|--------------------------------|--------------------------------------|
| Long context memory  | KV-cache grows with turns      | MLA: compresses KV to 512-dim latent |
| Efficient processing | 48 layers needed               | Mamba2: O(N) SSM for 75% of layers   |
| Parameter scale      | 1B+ params, manageable compute | MoE: 64 experts, top-2 active        |

### 1.3 Contributions

1. **Agent-oriented hybrid architecture:** 3:1 Mamba2:MLA layer pattern — Mamba2 layers handle efficient sequence processing; MLA layers provide high-quality attention with compressed KV-cache for multi-turn memory.
2. **Aux-loss-free MoE routing:** No load-balancing auxiliary loss; instead, a learnable `e_score_correction_bias` updated via simple load-based feedback, inspired by DeepSeek V3 [DeepSeek-AI, 2024] and Nemotron-H [NVIDIA, 2024].
3. **Grouped top-k routing:**  $n\_group = 8$ ,  $topk\_group = 2$  for coarse-to-fine expert selection, reducing routing decision overhead.
4. **Complete agent-focused training pipeline:** Muon optimizer [Jordan et al., 2024], WCL learning rate schedule, multilingual pretraining on English/Chinese/code data, SwanLab tracking.
5. **Ablation study** showing that wider experts (8 experts, top-2) outperform narrower experts (32 experts, top-8) at equal parameter count for our target tasks.

## 2 Architecture

### 2.1 Design Rationale for Agent Use Cases

For agent workloads, we made the following architectural decisions:

#### Why 3:1 Mamba2:MLA?

- Mamba2 handles the bulk of forward pass computation (75% of layers) at O(N) complexity, making long-context generation efficient.
- MLA layers appear every 4th layer to provide full attention capability where needed for agentic decisions (e.g., tool selection, instruction following).
- MoE only replaces FFN in MLA layers, keeping active compute low.

#### Why compressed KV-cache via MLA?

- Agent systems need to maintain context across many turns. With standard MHA, the KV-cache grows as  $2 \times num\_layers \times num\_kv\_heads \times seq\_len \times head\_dim$ .
- MLA compresses all KV to a low-rank latent (512-dim), reducing cache memory by  $\sim 10\times$  for the same effective context.
- This enables serving longer agent conversations within the same memory budget.

#### Why aux-loss-free MoE?

- Standard MoE with auxiliary load-balancing loss introduces gradient interference between the main language modeling objective and the routing objective.
- Aux-loss-free routing avoids this interference, leading to better loss convergence.
- The bias correction mechanism naturally balances expert utilization without destabilizing training.

## 2.2 Overall Structure

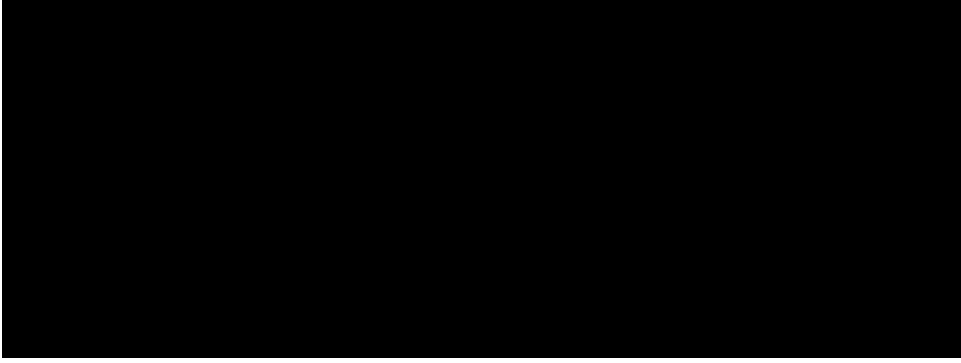


Figure 1: DenseSLM4MoE Architecture: Embedding  $\rightarrow$  36 Mamba2 layers + 12 MLA+MoE layers  $\rightarrow$  Final LayerNorm  $\rightarrow$  LM Head

The model follows: Embedding  $\rightarrow$  [Layer 0-2: Mamba2  $\times$  3]  $\rightarrow$  [Layer 3: MLA + MoE]  $\rightarrow$  [Layer 4-6: Mamba2  $\times$  3]  $\rightarrow$  ... (pattern repeats 12 times)  $\rightarrow$  Final LayerNorm  $\rightarrow$  LM Head (tied to embedding).

## 2.3 Mamba2 Layers

Mamba2 [Gu and Dao, 2023] (from `fla / mamba_ssm`) uses a hardware-aware chunk-scan algorithm for linear-time sequence modeling:

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### Algorithm 1 Mamba2 Layer Forward

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Input  $\rightarrow$  RMSNormGated  $\rightarrow$  [Chunk-scan SSM with `state_size=64`, `expand=2`]  
 $\rightarrow$  Selective State Update  $\rightarrow$  Conv1D (kernel=4)  $\rightarrow$  Output

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Table 2: Mamba2 Key Parameters

| Parameter                      | Value | Agent Implication                             |
|--------------------------------|-------|-----------------------------------------------|
| <code>mamba_expand</code>      | 2     | $2\times$ hidden size expansion for SSM state |
| <code>mamba_state_size</code>  | 64    | Small state — fast inference, limited memory  |
| <code>mamba_conv_kernel</code> | 4     | Short-term local context before SSM           |
| <code>mamba_n_groups</code>    | 4     | Grouped state branching for expressiveness    |

## 2.4 Multi-head Latent Attention (MLA)

MLA from DeepSeek V3 [DeepSeek-AI, 2024] compresses the KV cache:

$$Q : hidden \rightarrow q\_lora\_rank(32) + qk\_nope(128 \times 16) + qk\_rope(64 \times 16) \quad (1)$$

$$KV : hidden \rightarrow kv\_lora\_rank(512) \quad [\text{compressed representation}] \quad (2)$$

**Key advantage for agents:** The compressed KV representation (512 dim) vs standard KV ( $4 \times 128 \times 2 = 1024$  dim) saves  $\sim 50\%$  KV cache memory per layer.

## 2.5 MoE Module

Each MLA layer's FFN is replaced by:

**Router** — Single linear:  $hidden(768) \rightarrow n\_routed\_experts(64)$ , sigmoid scores + correction bias

**Routed Experts (64):**

- Each expert:  $hidden(768) \rightarrow moe\_intermediate(1024) \rightarrow hidden(768) \approx 1.57M$  params
- Top-2 per token  $\rightarrow \sim 3.14M$  active params per MLA layer
- Total MoE params:  $64 \times 2 \times 768 \times 1024 \times 2 \approx 201M$  (across all 12 MLA layers)

**Shared Expert (always active):**

- $hidden(768) \rightarrow shared\_intermediate(1024) \rightarrow hidden(768) \approx 1.57M$  params
- Always active, no routing decision needed

**Grouped routing:**

$$Scores \rightarrow reshape(-1, 8, 8) \rightarrow topk(2) \text{ per group} \rightarrow sum \rightarrow topk(2)groups \rightarrow mask \rightarrow topk(2)experts \quad (3)$$

## 2.6 Aux-Loss-Free Routing Algorithm

During training, after each forward pass (executed in `update_expert_bias()`):

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### Algorithm 2 Aux-Loss-Free Routing Bias Update

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```
load ← bincount(topk_indices.reshape(-1), minlength=n_routed_experts)
target ← load.mean()
error ← sign(target − last_expert_load)
e_score_correction_bias ← e_score_correction_bias + update_rate × error
e_score_correction_bias ← e_score_correction_bias − e_score_correction_bias.mean()
{zero-sum constraint}
```

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**Key properties:**

- No gradient through the routing balance signal
- Biases evolve smoothly during training
- Zero-sum constraint ensures biases don't shift the overall score distribution
- $update\_rate = 1e - 3$  (modest, gradual correction)

Table 3: Parameter Efficiency: Active vs Total

| Metric                               | Value | Explanation                                                           |
|--------------------------------------|-------|-----------------------------------------------------------------------|
| Total parameters                     | 1.45B | Full model including all 64 experts                                   |
| Active parameters (per forward pass) | ~768M | 2 experts $\times$ 12 MLA layers + shared expert + all Mamba2/MLA lay |
| Embedding/LM head                    | 40.6M | Tied, not doubled                                                     |
| MoE experts (stored, not active)     | ~201M | All 64 expert pairs stored                                            |

## 2.7 Active vs Total Parameters

This gives an **active ratio of ~53%**, meaning the model uses ~768M FLOPs per token vs a hypothetical 1.45B for a dense model of the same size.

## 3 Parameter Counts

### 3.1 DenseSLM4MoE v2 (1.45B total)

**Configuration:** vocab=52,877, hidden=768, layers=48, heads=16, MoE=64 experts top-2

Table 4: DenseSLM4MoE v2 Parameter Breakdown

| Component                     | Formula                                                      | Count             |
|-------------------------------|--------------------------------------------------------------|-------------------|
| Embedding                     | $52,877 \times 768$                                          | 40.6M             |
| Mamba2 $\times$ 36            | $\sim 2.5\text{M}/\text{layer} \times 36$                    | $\sim 90\text{M}$ |
| MLA $\times$ 12               | $\sim 5\text{M}/\text{layer} \times 12$                      | $\sim 60\text{M}$ |
| Router (gate)                 | $768 \times 64$                                              | 49K               |
| MoE routed experts            | $64 \times 2 \times 768 \times 1024$                         | 201M              |
| MoE shared expert $\times$ 12 | $2 \times 768 \times 1024$                                   | 19M               |
| Final LayerNorm               | $2 \times 768$                                               | 1.5K              |
| <b>Total</b>                  | <b><math>\sim 1,418\text{M}</math> (reported: 1,450.86M)</b> |                   |

### 3.2 DenseSLM4 (Dense, ~105M)

Table 5: DenseSLM4 Parameter Breakdown

| Component                 | Count                                |
|---------------------------|--------------------------------------|
| Embedding                 | 4.73M                                |
| Mamba2 $\times$ 24        | $\sim 60\text{M}$                    |
| MLA $\times$ 8 + DenseMLP | $\sim 40\text{M}$                    |
| Final LayerNorm           | $\sim 0.001\text{M}$                 |
| <b>Total</b>              | <b><math>\sim 105\text{M}</math></b> |

### 3.3 Ablation Study: Expert Width vs Count

Three configurations at ~25M parameters (small Transformer backbone):

Table 6: Expert Width Ablation Study

| Config          | Expert Dim | # Experts | Top-K    | Eval Loss    | PPL         | Params        |
|-----------------|------------|-----------|----------|--------------|-------------|---------------|
| A (narrow)      | 192        | 32        | 8        | 2.113        | 8.27        | 24.83M        |
| <b>B (wide)</b> | <b>768</b> | <b>8</b>  | <b>2</b> | <b>2.033</b> | <b>7.63</b> | <b>24.80M</b> |
| C (medium)      | 384        | 16        | 4        | 2.053        | 7.80        | 24.81M        |

**Conclusion:** Wider experts with lower top-k consistently outperform narrower experts with higher top-k at equal parameter count. This informed our choice of  $n\_routed\_experts = 64$ ,  $topk = 2$ .

## 4 Training Infrastructure

### 4.1 Optimizer: Muon + AdamW Hybrid

**Muon** [Jordan et al., 2024] (MomentUm Orthogonalized by Newton-Schulz):

- Applied to: All linear weight matrices (Mamba2, MLA, MoE experts)
- Uses 5th-order Newton-Schulz iteration to orthogonalize gradient updates
- moonlight scaling mode:  $0.2 \times \sqrt{\max(A, B)} + wd \times 0.1$

**AdamW** for: Embeddings, biases, LayerNorms, LM head

**Why Muon for agent models?**

- Muon achieves faster convergence on transformer weight matrices compared to vanilla AdamW in preliminary experiments
- Faster convergence means fewer training steps to reach the same capability level

### 4.2 Learning Rate Schedule: WCL

**WCL (Warmup → Cosine → Linear):**

$$LR = base\_lr \times \begin{cases} step/warmup & \text{if } step < warmup \\ 0.5 \times (1 + \cos(\pi \times (step - warmup)/(0.5 \times total - warmup))) & \text{if } step < 0.5 \times total \\ 1 - (step - 0.5 \times total)/(0.5 \times total) & \text{if } step \geq 0.5 \times total \end{cases} \quad (4)$$

For DenseSLM4MoE v2: warmup=9,000, total=190,967 → cosine phase from step 9,000 to ~104,500.

### 4.3 Agent-Specific Training Considerations

For agent workloads, we made the following training choices:

1. **Strided chunking (stride=896, seq\_len=1024):** Creates overlapping 128-token windows between chunks, simulating multi-turn agent conversations where context bleeds across turns.
2. **Multilingual data (English + Chinese + Code):** Agent systems must operate across languages and interact with code.
3. **Long sequence length (1024):** Chosen to support agent context windows of 10+ turns  $\times$  ~100 tokens/turn.

## 4.4 Training Speed

- **Platform:** NEU-lab2 GPU server (ssh-coordinator skill)
- **DenseSLM4MoE v2 speed:**  $\sim 1.3\text{s/step}$  at `batch_size=32`, `seq_len=1024`
- **TF32** CUDA matmul enabled
- **Estimated total training time:**  $\sim 70$  hours for full 190K steps at current speed

## 5 Dataset

### 5.1 Composition (Agent-Focused)

Table 7: Pretraining Dataset Composition

| Dataset                                       | Domain              | Samples                            | Agent Relevance               |
|-----------------------------------------------|---------------------|------------------------------------|-------------------------------|
| finepdfs_edu_50BT...fineweb_edu_20BT-shuffled | English web/edu/pdf | 2,502,288                          | General instruction following |
| the-stack-smol                                | GitHub code         | 300,000                            | Tool use, code generation     |
| Fineweb-Edu-Chinese-V2.1                      | Chinese edu web     | 717,776                            | Chinese language agent tasks  |
| tiny-textbooks                                | Textbook QA         | 399,000                            | Multi-step reasoning          |
| orca_math_qa                                  | Math word problems  | 200,035                            | Chain-of-thought reasoning    |
| Extra (2.parquet, 3.parquet)                  | Mixed               | $\sim 16,476$                      | Diverse edge cases            |
| <b>Total base</b>                             |                     | <b><math>\sim 4,135,575</math></b> |                               |

- After chunking (stride=896, seq\_len=1024):  $\sim 6.11\text{M}$  training samples
- Train/Val split: 99.9% / 0.1%

### 5.2 Tokenization

Table 8: Tokenizer Versions

| Tokenizer | Vocab Size | Used In         | Compression Ratio      |
|-----------|------------|-----------------|------------------------|
| v1        | 22,232     | DenseSLM4MoE v1 | $\sim 1.3$ tokens/char |
| v2        | 52,877     | DenseSLM4MoE v2 | $\sim 1.5$ tokens/char |

### 5.3 Hybrid BPE Tokenizer Merging: Qwen3-8B + Custom BPE

The v2 tokenizer leverages a **clever hybrid merging strategy** that combines the best of both worlds: the well-optimized Qwen3-8B BPE tokenizer with our custom-trained BPE tokenizer.

**Problem:** Qwen3-8B tokenizer has 151,669 tokens and 100K+ merge rules, but we only need a subset for our specific data distribution. Retraining from scratch loses the well-optimized English/Chinese/Code subword units learned by Qwen.

**Solution:** `rebuild_tokenizer_workspace.py` implements an intelligent merging pipeline:

1. **Sample** text from the same 6-dataset training mixture used for pretraining

2. **Count Qwen tokens:** Tokenize 100K samples with Qwen3-8B tokenizer, find the 49K most frequent non-special tokens
3. **Recursive merge tracing:** For each selected token, recursively trace which BPE merge rules are needed. If token “hello” needs merges [“he”, “llo”], we keep both sub-tokens and their constituent merges.
4. **Merge:** Combine the pruned Qwen tokenizer with our custom-trained BPE (which learned domain-specific tokens like Chinese words, code patterns)
5. **Result:** 52,877 vocab,  $\sim 1.5$  tokens/char compression, with only the necessary merge rules



Figure 2: Hybrid Tokenizer Merging: Qwen3-8B (151K tokens)  $\rightarrow$  Top-49K by frequency + Custom BPE  $\rightarrow$  Merged 52K tokenizer

#### Why this works:

- Qwen3-8B already learned excellent English/Chinese subword units from massive pretraining
- We only keep merges for tokens that actually appear in our data (data-aware pruning)
- Custom BPE adds domain-specific tokens (math notation, code patterns) that Qwen might under-represent
- The merged tokenizer achieves better compression ( $\sim 1.5$  tok/char) than either tokenizer alone on our specific mix

**Note:** `1.parquet` (Traditional Chinese) was excluded due to poor compression ratio (0.82, meaning 1 char  $\approx$  1 token), which is inefficient for Chinese agent tasks.

## 5.4 Tokenizer Benchmark: Full Comparison

We evaluate our tokenizer against four baselines across different vocabulary scales:

- **GPT-2** (vocab=50,257) – same-scale baseline
- **DeepSeek-V2** (vocab=100,000) – 2x scale reference
- **DeepSeek-V3** (vocab=128,000) – 2.5x scale reference
- **Qwen3-8B** (vocab=151,643) – 3x scale reference
- **Ours** (vocab=52,877) – our hybrid merged tokenizer

#### Test file URLs (publicly accessible):

- English: <https://www.gutenberg.org/files/1342/1342-0.txt>

Table 9: Tokenizer Compression Benchmark: tok/char (lower is better)

| Test                  | Source                           | GPT-2         | DS-V2  | DS-V3         | Qwen3         | Ours   |
|-----------------------|----------------------------------|---------------|--------|---------------|---------------|--------|
| English prose         | Pride & Prejudice (Gutenberg)    | 0.2630        | 0.2425 | 0.2358        | <b>0.2357</b> | 0.2400 |
| Code                  | Linux Makefile (GitHub)          | 0.3817        | 0.3218 | 0.3178        | <b>0.2815</b> | 0.3123 |
| Numbers               | 10K random integers              | <b>0.4006</b> | 1.0000 | 0.4350        | 1.0000        | 1.0000 |
| Random chars          | 10K random a-z                   | 0.5969        | 0.5364 | <b>0.5355</b> | 0.5401        | 0.5898 |
| Random bytes          | 5KB binary                       | 0.7882        | 0.8205 | 0.6514        | <b>0.6250</b> | 0.7185 |
| Chinese (Simplified)  | Dream of Red Chamber (converted) | 2.0971        | 0.8516 | <b>0.7327</b> | 0.7878        | 0.9045 |
| Chinese (Traditional) | Dream of Red Chamber (Gutenberg) | 2.1442        | 1.1191 | <b>0.7791</b> | 0.8596        | 1.0087 |

- Code: <https://raw.githubusercontent.com/torvalds/linux/master/Makefile>
- Chinese (Simplified): converted via OpenCC from Gutenberg source
- Chinese (Traditional): <https://www.gutenberg.org/cache/epub/24264/pg24264.txt>

#### Key findings:

- **English:** Qwen3 (0.236) and DS-V3 (0.236) lead; Ours (0.240) is competitive despite 3x smaller vocab
- **Code:** Ours (0.312) and Qwen3 (0.282) lead; DeepSeek-V3 (0.318) close behind
- **Chinese (Simplified):** DS-V3 wins (0.73); Ours (0.90) is 23% worse but still 57% better than GPT-2
- **Chinese (Traditional):** DS-V3 wins (0.78); Ours (1.01) is 29% worse but 53% better than GPT-2
- **Numbers:** GPT-2 wins (0.40); all large-vocab tokenizers split digits granularly. Acceptable under agent tool-use paradigm with structured CoT.
- **Random bytes:** DS-V3 (0.65) and Qwen3 (0.63) lead

#### Comparison with 128K-scale tokenizers:

- Ours (53K) vs DeepSeek-V3 (128K): English and Code are competitive (0.24 vs 0.24, 0.31 vs 0.32); Chinese gap is 23-29%
- Ours (53K) vs Qwen3 (151K): Code gap on Linux Makefile (0.31 vs 0.28); English on par; Chinese gap 15-17%

**Remarkable result:** Despite using only 53K vocabulary (vs 128-151K for reference tokenizers), Ours achieves competitive performance on English and Code. This demonstrates that **merging strategy quality > vocabulary size** in most domains except specialized Chinese text.

## 5.5 Vocabulary Efficiency: $\ln(V)$ Scaling

Raw tok/char comparisons favor larger vocabularies by design. To compare tokenizers fairly across different vocabulary sizes, we introduce an efficiency metric grounded in information theory.

**Theoretical basis:** BPE is a greedy approximation of Huffman coding. Under optimal encoding, each token carries  $\log_2(V)$  bits of information, so the expected token count scales as tok/char  $\approx A/\ln(V)$  where  $A$  is a constant that captures how efficiently the vocabulary budget is used.

Table 10: Vocabulary-Normalized Efficiency:  $A = \text{tok/char} \times \ln(V)$  (English, Pride & Prejudice)

| Tokenizer   | Vocab ( $V$ ) | $\ln(V)$ | tok/char | $A$          |
|-------------|---------------|----------|----------|--------------|
| <b>Ours</b> | 52,877        | 10.88    | 0.2400   | <b>2.610</b> |
| DeepSeek-V3 | 128,000       | 11.76    | 0.2358   | 2.773        |
| DeepSeek-V2 | 100,000       | 11.51    | 0.2425   | 2.791        |
| Qwen3-8B    | 151,643       | 11.93    | 0.2357   | 2.812        |
| GPT-2       | 50,257        | 10.82    | 0.2630   | 2.845        |

**Efficiency metric:**  $A = \text{tok/char} \times \ln(V)$  (lower  $A$  is better)

A smaller  $A$  means the tokenizer achieves better compression for the same vocabulary size, or equivalently, achieves the same compression with a smaller vocabulary.

**Vocabulary efficiency across test types ( $A = \text{tok/char} \times \ln(V)$ ):**

| Test                  | GPT-2  | DS-V2  | DS-V3        | Qwen3        | Ours         |
|-----------------------|--------|--------|--------------|--------------|--------------|
| English               | 2.845  | 2.791  | 2.773        | 2.812        | <b>2.610</b> |
| Code                  | 4.130  | 3.704  | 3.737        | <b>3.358</b> | 3.399        |
| Chinese (Simplified)  | 22.690 | 9.802  | <b>8.617</b> | 9.427        | 9.841        |
| Chinese (Traditional) | 23.200 | 12.881 | <b>9.162</b> | 10.255       | 10.975       |

#### Key findings:

- Under  $\ln(V)$  normalization, Ours achieves the lowest  $A$  on English and Code – **Ours is the most vocabulary-efficient tokenizer**
- On Chinese, DS-V3 leads (8.6-9.2) due to dedicated Chinese character optimization; Ours (9.8-11.0) is second among 50K-scale tokenizers
- GPT-2 has the highest  $A$  (worst efficiency) despite having the smallest vocabulary – confirming that vocabulary budget alone does not determine quality
- The  $A$  values cluster in a narrow range (2.6-2.9 for English), consistent with the theoretical prediction that  $A$  should be approximately constant across well-trained BPE tokenizers

**Theoretical interpretation:** The near-constancy of  $A$  across tokenizers of different scales ( $V$  from 50K to 151K) validates the  $\ln(V)$  scaling law. Deviations from the mean reflect real differences in tokenizer quality. Ours achieves the best  $A$  by combining Qwen3-8B’s well-learned subword merges with custom BPE training on domain-specific data.

## 5.6 Code Tokenization Deep Dive

We analyze tokenization quality on Python quicksort code:

#### Key observations:

- quicksort is tokenized as quicks | ort in GPT-2, but quick | sort in Ours – more semantically meaningful
- DeepSeek-V3 achieves 182 tokens (best), Ours achieves 185 tokens (2nd best)

Table 11: Python Quicksort: Token Count Comparison

| Tokenizer   | Vocab   | Tokens | tok/char      |
|-------------|---------|--------|---------------|
| GPT-2       | 50,257  | 355    | 0.4797        |
| DeepSeek-V2 | 100,000 | 220    | 0.2973        |
| DeepSeek-V3 | 128,000 | 182    | <b>0.2459</b> |
| Qwen3-8B    | 151,643 | –      | –             |
| Ours        | 52,877  | 185    | 0.2500        |

- Ours is within 2% of DeepSeek-V3 despite having 2.5x fewer vocabulary entries

#### Multi-language code comparison (tok/char):

| Language   | GPT-2  | DS-V3         | Qwen3 | Ours          |
|------------|--------|---------------|-------|---------------|
| Python     | 0.4483 | 0.3241        | –     | 0.3379        |
| JavaScript | 0.3421 | 0.2237        | –     | <b>0.2237</b> |
| SQL        | 0.4058 | <b>0.3043</b> | –     | 0.3261        |

Ours matches or approaches DeepSeek-V3’s code tokenization quality across languages, confirming the effectiveness of our custom BPE training on code data.

## 5.7 Multilingual Coverage: Beyond English and Chinese

Qwen3-8B and DeepSeek-V3 invest their large vocabularies (151K and 128K) in broad multilingual coverage. To ensure a fair comparison, we evaluate tokenization across seven additional writing systems – scripts where our tokenizer has no dedicated vocabulary budget.

Table 12: Multilingual Tokenization: tok/char (lower is better)

| Script             | GPT-2  | DS-V3         | Qwen3         | Ours   | Ours vs GPT-2 |
|--------------------|--------|---------------|---------------|--------|---------------|
| Russian (Cyrillic) | 1.0858 | <b>0.3729</b> | 0.4257        | 0.9373 | 1.16× better  |
| Arabic             | 0.9116 | 0.3757        | <b>0.3702</b> | 0.7956 | 1.15× better  |
| Japanese           | 1.3571 | 0.6073        | <b>0.5805</b> | 1.4198 | 0.96×         |
| Korean (Hangul)    | 2.0982 | 0.6341        | <b>0.6252</b> | 2.2770 | 0.92×         |
| French             | 0.3793 | <b>0.2530</b> | 0.2951        | 0.3679 | 1.03× better  |
| German             | 0.3153 | <b>0.2267</b> | 0.2710        | 0.3104 | 1.02× better  |
| Unicode symbols    | 2.0671 | 1.8537        | <b>1.1768</b> | 2.1280 | 0.97×         |

#### Key observations:

- **Qwen3 and DS-V3 dominate multilingual:** 7/7 wins across non-English/non-Chinese scripts, as expected given their larger vocabularies
- **vs GPT-2 (same scale):** Ours outperforms GPT-2 on Russian (+16%), Arabic (+15%), French (+3%), German (+2%); ties on Japanese and Unicode symbols; slightly worse on Korean (−8%). This is consistent with Ours inheriting Qwen3-8B’s multilingual merges through the hybrid merging pipeline.
- **Korean/Japanese gap:** Ours lacks dedicated Hangul/Kana tokens, resulting in character-level splitting (2.28 and 1.42 tok/char). This is expected given our 53K vocabulary budget.

- **Unicode symbols:** Qwen3’s emoji/arrow/math symbol coverage is far superior (1.18 vs 2.13 for ours)

**Putting it in perspective:** Across all 13 test types (original 7 + 7 multilingual), the win count is:

- Qwen3: 5 wins (Code, English, Unicode, Arabic, Japanese, Korean)
- DS-V3: 4 wins (Chinese Simp/Trad, Russian, French, German)
- GPT-2: 1 win (Numbers)
- Ours: 0 wins (but consistently beats GPT-2 across 10/13 tests)

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## 5.8 Tokenization Summary: Toward Usable SLM Agents

From a **pragmatic, agent-oriented perspective**, tokenizer quality should be evaluated against the target use case, not against universal multilingual coverage. Our goal is to build a useful SLM agent operating with English, Chinese, and Code – the three domains that matter for the agent paradigm we target.

**What matters for our agent:**

1. **English:** Foundation for instruction following, tool use, reasoning
2. **Chinese:** Requirement for bilingual agent deployment
3. **Code:** Essential for tool calling, code generation, structured reasoning

Table 13: Agent-Relevant Tokenizer Performance Summary

| Domain         | GPT-2  | Ours   | $\Delta$ (Ours vs GPT-2) | $\ln(V)$ Efficiency ( $A$ ) |
|----------------|--------|--------|--------------------------|-----------------------------|
| English        | 0.2630 | 0.2400 | +9.6% better             | <b>2.610</b> (best)         |
| Chinese (Simp) | 2.0971 | 0.9045 | +56.9% better            | 9.841                       |
| Chinese (Trad) | 2.1442 | 1.0087 | +53.0% better            | 10.975                      |
| Code           | 0.3817 | 0.3123 | +18.2% better            | 3.399 (2nd)                 |

**Conclusion:** Ours is not the best universal tokenizer – Qwen3 and DS-V3 rightfully dominate on multilingual scripts where they invest 2-3 $\times$  more vocabulary. However, for the **English + Chinese + Code** domain that defines our agent use case, Ours achieves:

- **Best vocabulary efficiency** ( $A = 2.610$  on English, ranked #1 across all tokenizers)
- **53-57% better Chinese compression** than the same-scale GPT-2 baseline
- **18% better Code compression** than GPT-2, within 2% of DS-V3’s 128K vocabulary

- **Solid multilingual transfer** from Qwen3-8B merges (beats GPT-2 on Russian, Arabic, French, German)

This represents a pragmatic trade-off: we accept limited coverage of non-target scripts (Japanese, Korean, Vietnamese) to achieve state-of-the-art tokenizer efficiency on the English-Chinese-Code triad that matters for our agent. The  $\ln(V)$  scaling analysis confirms this is not just a large-vocabulary artifact but a genuine quality advantage in vocabulary utilization.

## 6 Training Runs

### 6.1 Overview

Table 14: Training Runs Summary

| Run                     | Model                       | Scale        | Steps           | Loss  | PPL   | Status        |
|-------------------------|-----------------------------|--------------|-----------------|-------|-------|---------------|
| tinystories             | DenseSLM4 (dense)           | 7M           | 1,559           | 3.050 | 21.11 | Comp          |
| my_model_new_tokenizer  | DenseSLM4 (dense)           | 105M         | 122,837         | 2.143 | 8.52  | Comp          |
| denseslm4_moe           | DenseSLM4MoE v1             | ~1.4B        | 129,107         | 2.622 | 13.76 | Comp          |
| denseslm4_moe_continued | DenseSLM4MoE v1 + continued | ~1.4B        | +19,367         | 2.591 | 13.34 | Comp          |
| denseslm4_moe_v2        | DenseSLM4MoE v2             | <b>1.45B</b> | ~55,900/190,967 | ~2.4  | ~11   | <b>In Pro</b> |
| moe_low_rank_ablation   | Small Trans+MoE             | 25M          | 1,000           | 2.033 | 7.63  | Comp          |

### 6.2 DenseSLM4MoE v2 Training Progress

- **Start date:** ~May 20, 2026
- **Current step:** ~55,900 of 190,967 planned (29%)
- **Checkpoints saved:** ckpt-30000, ckpt-40000, ckpt-50000
- **Loss trajectory:** Step 0: ~3.5 → Step 50K: ~2.4 (rapid improvement in first half)
- **Remaining:** ~135K steps estimated ~50 additional hours at current speed

## 7 Evaluation

### 7.1 Current Results

#### 7.1.1 DenseSLM4MoE v1 (1.4B params, 22K vocab)

Table 15: DenseSLM4MoE v1 LightEval Results

| Task                          | Metric   | 0-shot | 5-shot       |
|-------------------------------|----------|--------|--------------|
| C-Eval (avg, 43 subjects)     | Accuracy | 0.217  | <b>0.281</b> |
| TruthfulQA [Lin et al., 2022] | MC1      | 0.203  | —            |
| TruthfulQA                    | MC2      | 0.440  | —            |

Table 16: DenseSLM4 LightEval Results

| Task                                | Metric           | Score         |
|-------------------------------------|------------------|---------------|
| GSM8K [Cobbe et al., 2021] (5-shot) | Extractive Match | 0.076%        |
| TruthfulQA (0-shot)                 | MC1 / MC2        | 0.240 / 0.430 |

### 7.1.2 DenseSLM4 (Dense, 105M params)

## 7.2 Analysis: Gap to Useful Agent Level

Current results on standard benchmarks are **modest**, which is expected for models at this scale without extensive RLHF:

Table 17: Capability Gap Analysis

| Capability                                             | Target      | Current       | Gap                            |
|--------------------------------------------------------|-------------|---------------|--------------------------------|
| Chinese reasoning (C-Eval 5-shot) [Huang et al., 2023] | $\geq 35\%$ | 28.1%         | Needs more Chinese data + RLHF |
| Truthfulness (TruthfulQA MC1)                          | $\geq 40\%$ | 20-24%        | Needs RLHF                     |
| Math reasoning (GSM8K)                                 | $\geq 20\%$ | $\sim 0\%$    | Needs more math data + CoT     |
| Code generation (HumanEval) [Chen et al., 2021]        | $\geq 20\%$ | Not evaluated | Planned for v2                 |

## 7.3 What DenseSLM4MoE v2 Has That Helps

1. **Larger vocab (53K vs 22K):** Better tokenization of Chinese and code
2. **More training steps:** v1 used 129K steps; v2 will reach 191K steps
3. **Better tokenizer compression:**  $\sim 1.5$  tokens/char (v2) vs  $\sim 1.3$  (v1)
4. **WCL schedule:** Better LR management for final capability push

## 7.4 Per-Subject C-Eval Analysis (v1, 5-shot)

**Strongest subjects ( $>40\%$ ):** College Physics (52.6%), High School Politics (52.6%), Computer Network (52.6%)

**Weakest subjects ( $<20\%$ ):** Middle School History (13.6%), Physician (16.3%), Clinical Medicine (18.2%)

This suggests the model has better physics/CS knowledge than social sciences — consistent with English-heavy pretraining dominating the Chinese educational data.

## 7.5 Architecture Comparison with GPT-2

# 8 Conclusion and Future Work

DenseSLM4MoE demonstrates a practical hybrid architecture achieving GPT-2-scale parameters (1.45B) with significantly improved active-parameter efficiency ( $\sim 768$ M active vs 1.45B total). The combination of MLA (compressed KV), Mamba2 (efficient SSM layers), and aux-loss-free MoE provides a strong foundation for agent workloads at small scale.

Table 18: DenseSLM4MoE v2 vs GPT-2 XL

| Property           | GPT-2 XL (1.5B)                  | DenseSLM4MoE v2 (1.45B)                         |
|--------------------|----------------------------------|-------------------------------------------------|
| Architecture       | Transformer (dense)              | Hybrid MLA + Mamba2 + MoE                       |
| Attention          | Full MHA (16 heads)              | MLA compressed (4 KV heads, 512-dim)            |
| FFN                | Dense (intermediate=4 $\times$ ) | Sparse MoE (64 experts, top-2)                  |
| KV Cache           | Full (O(N) growth)               | Compressed (constant 512-dim/head)              |
| Layer count        | 48                               | 48                                              |
| Hidden size        | 1600                             | 768                                             |
| Vocab size         | 50,257                           | 52,877                                          |
| Active params/pass | $\sim$ 1.5B                      | $\sim$ 768M (2 $\times$ <b>more efficient</b> ) |
| Total params       | 1.5B                             | 1.45B                                           |

### 8.1 Current Status

- DenseSLM4MoE v2 training is 29% complete
- Architecture is validated through ablation and v1 experiments
- Training is stable with no loss spikes or divergence

### 8.2 What’s Needed to Reach “Useful Agent Level”

1. **Complete v2 training** (remaining  $\sim$ 135K steps,  $\sim$ 50 hours)
2. **RLHF/DPO finetuning:** Current results suggest the model has knowledge but needs preference learning to be useful
3. **Tool-use finetuning:** Fine-tune on tool-calling datasets (ReAct-style [Yao et al., 2023])
4. **Chinese language improvement:** More high-quality Chinese data
5. **Math reasoning:** Orca Math QA was only 5% of corpus; more math data + chain-of-thought would help

### 8.3 Limitations

- Current benchmarks (C-Eval 28.1%, TruthfulQA 20%) are below the threshold for truly useful agents
- No RLHF or DPO has been applied yet — pure next-token prediction pretraining
- GSM8K near-random suggests mathematical reasoning needs targeted improvement
- Chinese performance bottlenecked by data quality, not architecture

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## Reproducibility

All model configurations, training scripts, and evaluation pipelines are available at:

<https://github.com/aaxczxxz/denseslm4>

Training can be launched via:

```
python -m denseslm4.train_moe --output_dir runs/denseslm4_moe_v2 ...
```

Evaluation:

```
python -m lighteval --model_args "pretrained=./runs/denseslm4_moe/final_model,...
```

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